SHAHRAM AJLOO

Process Lead · design for assembly, process validation and serial production

+45 93 99 95 14 · ajlooshahram@gmail.com · linkedin.com/in/shahramajloo · Copenhagen area

4 November 2026

Process Lead, Nacelle Components · Technology & Operations · Vestas · Munkebo / Szczecin

Application: Process Lead, Nacelle Components

Dear Hiring Manager,

The assembly decision is taken about three years before the first nacelle is assembled. A bracket that needs three hands and a mirror, a harness routed through a gap that closes when the next component arrives, a hydraulic connection that can only be reached from below — none of that is a production problem. It is a design decision that was approved without the person who has to build it in the room. That is why the manufacturing interface to **R&D** is the part of this role I want most: the cost of a change is measured in drawing revisions in year one and in stopped stations in year three.

First link: DfA, before the drawing is frozen. Design for Assembly is not a review at the end of a project; it is a set of questions asked while the geometry can still move: how many parts, how many orientations, how many hands, what does the operator see, and what happens when the tolerance stack goes the wrong way. I have reviewed drawings, specifications and coordination models for constructability and maintenance access on **CAPEX** builds, and I have been the one who had to install what someone else drew. That is where the credibility comes from when the answer is 'this will not assemble'.

Second link: DFMEA and PFMEA, as one conversation. A failure mode that engineering rates low and production sees every week is a rating problem, not a disagreement. So the **DFMEA** and the **PFMEA** are read against each other: what can fail in the product, what can fail in the process step that builds it, and which control actually detects it before the part leaves the station. I have run that discipline under the Machinery Directive and in GMP-regulated production, where a risk assessment without a control and an owner is a document, not a control.

Third link: control plan, parameters and work instruction. A **control plan** is a promise about where the measurement happens and what the operator does when it is out of limits — not a list of characteristics. Every **process parameter** has a value, a tolerance and a reason, and every **work instruction** is written so the person at the station can follow it without asking: what to pick, what to check, what to do if it is wrong. I have written protocols and instructions that were used at the bench rather than filed, including inspection and test plans with named hold points.

Fourth link: process validation and readiness. A process is proven on a defined number of units against a criterion written before the build, and the result is either met or not met — never 'mostly'. **FAT/SAT** and **IQ/OQ/PQ** on 37 projects taught me the difference between a test that shows the system runs and a test that shows it behaves: sequences, failure simulation, calibration checked before the reading, and **zero critical findings** at handover. That is the same logic as production validation, applied to an assembly station instead of a plant.

Fifth link: change control, and the maturity that follows it. A change to a process is a change to the product, and it needs one owner, one revision and one approval before it reaches the line. I have worked **change control** in SAP under GMP/FDA/EMA, where an unassessed change is a **non-conformance** waiting to happen, and I have carried the consequence of a change that arrived late: rework, a stopped station and a customer who asks why. **Process maturity** is what you can show an auditor and a new operator on the same day.

Technical leadership without line management. The strongest process engineers I have worked with were challenged on data, not on title. My part is to make the challenge concrete: here is the cycle time, here is the scrap, here is the station where the **containment** keeps coming back — and here is the change I would test on twenty units. I have coached apprentices and technicians with an individual plan, run **MUS** and built competence matrices, and I know that authority in a matrix organisation is earned by being right often enough, with evidence.

Practicalities. Based near Copenhagen; **Munkebo** is a domestic move and **Szczecin** is a commute I would take on a rotation, and either is fine — travel up to 50 days a year is what the role needs, not a constraint. Driving licence B. English professional, Danish native; used to working in a global team where the meeting is in English and the shop floor is not. Available at short notice, and I have read that applications close on 26 November.

Kind regards

Shahram Ajloo

SHAHRAM AJLOO

Process Lead · design for assembly, process validation and serial production

+45 93 99 95 14 · ajlooshahram@gmail.com · linkedin.com/in/shahramajloo · Copenhagen area

PROFILE

Manufacturing and process engineer from production and critical infrastructure: authorised marine and mechanical engineer (maskinmester) at LV/MV/HV level, electrical and mechanical design (EQF 6), B.Tech in technology management. Twelve years leading production and workshop teams; 37 CAPEX builds validated with **FAT/SAT**, IQ/OQ/PQ and **zero critical findings**; process documentation, change control and **SAP** in GMP-regulated manufacturing.

DESIGN FOR ASSEMBLY AND PROCESS DEVELOPMENT

- **Manufacturability reviews:** Drawings, specifications and coordination models reviewed for manufacturability, clearances and maintenance access before the first unit was built — and the same review repeated when the design changed.
- **Assembly sequence and interfaces:** Sequence, tooling and access planned around the real interfaces: electrical connections, wire harnessing, hydraulics, controls and the trade that comes next.
- **Work instructions that get used:** Work instructions and inspection sheets written with the people who execute them, tested on the shift with least support, and revised when a non-conformance shows they were unclear.

VALIDATION, CONTROL AND CHANGE

- **Process validation:** Criterion written before the build, units defined, result recorded while the system runs: FAT/SAT, IQ/OQ/PQ and functional testing with calibration checked first.
- **Control plans and parameters:** Control plan as a reaction plan per measurement — value, tolerance, reason and what the operator does when it is outside. Parameters changed one at a time with a before and after.
- **Change control:** Change control in SAP under GMP/FDA/EMA: one revision, one owner, one approval, and a reassessment of risk before the change reaches the line.

TECHNICAL LEADERSHIP AND PROBLEM SOLVING

- **Root cause into standard:** A3, 5 Whys and Ishikawa to the root cause, then a change to the standard — fixture, instruction, control or parameter — re-measured after 60 days. Standardisation and continuous improvement judged by whether the number moved.
- **Coaching without a title:** Process and workshop teams coached on the floor; individual plans, competence matrices and coaching that made the next decision possible without me.
- **Stakeholders and data:** Engineering, procurement, quality and operations held to the same numbers; the argument won with cycle time, scrap and deviation count rather than opinion.

EXPERIENCE

Owner / Technical Director — ELOHVAC ApS · Copenhagen · May 2023–present

Scope: 5 electricians + 3 apprentices · full P&L · design, build, validate, hand over

- **Process, cost and readiness:** Every job designed, built and validated against a criterion written first; the cost of getting it wrong was carried on my own P&L.

Senior Commissioning Engineer / Project Manager — Novo Nordisk A/S · Kalundborg / Bagsværd · Nov 2022–Jul 2024

Scope: 37 concurrent CAPEX projects · 25 specialists · GMP/FDA/EMA

- **Validation and documentation:** Protocols before the test, deviations and change control in SAP, and zero critical findings at handover — with the authority in the room.

Project and Workshop Manager — Ventek-Air · Copenhagen · Jan 2022–Nov 2022

Scope: 14 employees · 11 concurrent contracts · own workshop and assembly

- **Assembly under real constraints:** Components cut, formed and assembled in-house, then installed on 11 sites in parallel: sequence, tooling and access decided before the crew arrived.

Maintenance and Operations Manager — Biofos (Lynetten) · Copenhagen · Aug 2017–Jan 2022

Scope: Treatment plant in 24/7 operation · pumps, hydraulics, HVAC, HV switchgear, SCADA

- **Reliability as a process:** Preventive maintenance and planned outages with the fallback designed first; recurring faults driven to root cause, 40 per cent lower energy consumption.

Electromechanical Technician / Production Team Leader — ARJ Engineering · Tehran · 2003–2010 · 12 mechanics and technicians

EDUCATION AND AUTHORISATION

Authorised Marine and Mechanical Engineer — Danish electrical authorisation LV/MV/HV · Danish Safety Technology Authority · May 2023

B.Tech, Technology Management and Mechanical Engineering (EQF 6) · Copenhagen School of Marine Engineering & Technology Management · 2017–2022 · GPA 7.6/12

Electrical and Mechanical Design, Electrical Engineering (EQF 6) · Tehran Technical University · 2000–2003

Refrigeration and HVAC (AMU) · Lean: 5S, PDCA, VSM, A3 · Mercantec · 2021 · STCW safety certificates (2021–2023)

LANGUAGES, PRACTICALITIES AND REFERENCES

English: professional, written and spoken · **Danish:** native · **Persian:** native · **Based near Copenhagen** — Munkebo as a domestic move, Szczecin on rotation · **Travel:** up to 50 days a year is expected, not a problem · **Driving licence B · Available at short notice** · References on request.

FIRST 90 DAYS — FROM THE PROCESS STEP TO A CONTROL PLAN THAT HOLDS

Period	What happens, and how it is measured
Week 1–2	Walk the assigned process step with the operators who run it; read the current PFMEA, control plan, process flow and work instructions against what actually happens at the station, and list every point where the two disagree.
Day 30	One page per assembly step: sequence, interfaces (electrical connections, wire harnessing, hydraulics), takt and the control that detects a failure before the part moves on.
Day 60	First DfA input delivered to R&D for the next design gate, with cost and cycle-time consequence per proposal — and the first process change governed through change control with a documented re-validation.
Day 90	Validation of one process step against a written criterion, a revised work instruction proven with new operators, and a maturity gap list agreed with the Manufacturing Lead and the factory.

A PROCESS THAT ONLY WORKS FOR THE BEST OPERATOR IS NOT A PROCESS

The test of an assembly step is not the cycle time of the person who has done it for four years. It is the cycle time, the error rate and the traceability of the person in week two — and of the shift that works Friday afternoon. So the instruction, the fixture and the control are designed for that person, and the validation is run with them, not with the specialist. If the station needs a particular pair of hands, the process is not finished; it is dependent.

A DEFECT THAT CANNOT BE DETECTED AT THE STATION WILL BE DETECTED BY THE CUSTOMER

Every process step has at least one failure nobody can see once the part is closed, torqued or connected. That is exactly where the control has to sit — before the next component hides it. It is why I read the **PFMEA** backwards from the field: what is the worst thing this station can send on, and what measurement, poke-yoke or traceability catches it here. Where nothing can catch it, the design has to change, and that conversation belongs in **DfA** while it is still cheap.

CHALLENGE WITH NUMBERS, OR DO NOT CHALLENGE

Manufacturing knowledge is easy to have and hard to prove. The **data** is the proof: the cycle time at station four, the scrap on one family of brackets, the three containments in a month at the same step. With those numbers an engineer in production can change a design decision made by someone with a better title — and without them, the discussion becomes a matter of opinion, which the organisation will always resolve in favour of the drawing.

ONE MORE THING ABOUT THE ASSEMBLY LINE

A nacelle is not difficult to assemble because it is heavy. It is difficult because every step hands over to the next one through an interface that somebody designed — cables, harnesses, hydraulics, torque, access — and each of those interfaces is a decision that can be made well three years early or badly three years late. That is the work I am applying for: to be the person in the design room who knows what the station looks like, and the person on the station who knows why the drawing looks the way it does.

WHERE I WOULD START

At the station where the same **non-conformance** keeps coming back. First week: the process step I am given, its operators, its current control plan and its last hundred deviations — read together, not separately. Then the next design gate: the three things in the current design that cost the most to assemble, with the number that shows it, so the conversation with **R&D** starts with evidence rather than a wish.

WHAT THE ROLE REQUIRES — WHAT I HAVE DELIVERED	
Requirement in the posting	Evidence
Own end-to-end DfA and process development from early product development through industrialisation to serial production	Specification to validated handover on 37 CAPEX builds and 11 contracts: designed with the builder in the room, proven before serial use.
Act as the key manufacturing interface with R&D, shaping design through DfA and DFMEA	Manufacturability reviews of drawings, specifications and models — clearances, assembly and maintenance access — with the change proposed before the drawing froze.
Own and improve global process standards and deliverables: PFMEA, control plans, process flows, work instructions, process parameters	Control plans, inspection and test protocols and work instructions written and maintained; risk assessment linked to a control and an owner, not filed.
Define manufacturing concepts, assembly sequences, line balancing, tooling and production methods against safety, quality, cost and takt time	Sequence, tooling and access planned for 11 parallel sites and a 14-person workshop; capacity and takt time balanced against the plan weekly.
Govern manufacturing process changes and ensure technical compliance, process maturity and readiness across the lifecycle	Change control in SAP under GMP/FDA/EMA: one revision, one owner, one approval, and risk reassessed before the change reached the line.
Support Manufacturing Leads and factories through industrialisation, prototype and early-build activities	Early builds and first-of-a-kind installations supported on site — 37 projects, 25 specialists, issues solved where they occurred, with the fallback designed first.
Provide technical leadership and coaching to process engineers and designers without direct people-management responsibility	Technical lead in matrix projects: apprentices, technicians and specialists coached with an individual plan; authority built on evidence, not on title.
Bachelor's degree or equivalent in Mechanical, Electrical, Mechatronics Engineering or another relevant discipline	Electrical and mechanical design (EQF 6), B.Tech in technology management and mechanical engineering (EQF 6), authorised marine and mechanical engineer.
Significant experience in manufacturing engineering, process engineering or industrialisation, 8 years or more	Twenty years in technical project and production environments, twelve of them leading production and workshop teams of 5 to 14 people.
Solid experience with manufacturing and assembly processes, production systems and industrialisation of new products	Assembly of electrical panels, harnessing, pipework, hydraulics and equipment into operating systems — from first unit to repeatable handover, in regulated production.
Practical experience with PFMEA, control plans, process validation, process documentation, Windchill / Creo and change management	PFMEA logic through risk assessment (Machinery Directive, GMP); process validation through FAT/SAT and IQ/OQ/PQ; change in SAP — the PLM layer is the same discipline in Creo and Windchill.
Support prototype, early-build and NPI activities through to manufacturing readiness	NPI of equipment and installations: prototype and first build, readiness review, then handover to operations with documentation an operator can use on day one.
Experience from an international manufacturing environment: automotive, aviation, heavy engineering, medical manufacturing or wind energy	Heavy engineering and medical manufacturing: 37 CAPEX builds under GMP/FDA/EMA, 11 contracts with international suppliers.
Technical ownership and accountability for process performance, quality, safety, cost and manufacturing readiness	Full P&L since 2023: performance, quality, safety and cost owned on my own balance sheet, with the consequences of a bad process paid by me.
Analytical, structured problem-solving: root causes turned into sustainable improvements, and challenging existing solutions with data	A3, 5 Whys and Ishikawa to the root cause, a change to the standard, then a re-measurement after 60 days; designs challenged with data — cycle time, scrap, deviation count — before installation, not after.
Collaboration and stakeholder management across R&D, Manufacturing, Quality and global functions, and clear communication	Stakeholders in engineering, procurement, quality and operations on 37 projects and 11 contracts; global suppliers met in English, reports written for whoever was not in the room.